

LUIS ORLANDO SOLIS

Platform Engineering · DevSecOps · Site Reliability Engineering

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Executive Summary

What began as an independent infrastructure initiative has grown into eleven platform projects published on GitHub at github.com/luisorlandosolis — including one running in a live production environment and one demonstrating enterprise-grade Kubernetes platform engineering with a full CI/CD delivery layer. Built through self-directed learning, hands-on implementation, and troubleshooting real problems as they came up — including recovering a full hybrid environment from a 3-day power outage in 2.5 hours with no runbook and no outside help. Each platform is documented, version-controlled, and available to review on GitHub.

The work has moved through secure access, automation, provisioning, observability, security engineering, infrastructure recovery, centralized administration, backup architecture, a live production environment, Kubernetes platform engineering, and a full CI/CD delivery platform.

Portfolio Development Path

- Phase 1 — Foundation: hybrid infrastructure established — Hyper-V, Active Directory, Fedora, WireGuard mesh, Azure VM, Proxmox
- Phase 2 — Platform Engineering: 3-node HA cluster, custom weather app, Ansible automation, Grafana + FinOps monitoring
- Phase 3 — DevSecOps: internal PKI, mTLS, Fleet MDM across 10 hosts, Security Command Center, webhook-driven automation
- Phase 4 — SRE Validated: full incident recovery from a multi-day outage, GitHub publication, platform standardization
- Phase 5 — Portfolio Expansion: centralized enterprise administration, multi-tier backup & recovery, live production deployment
- Phase 6 — Cloud Native: 3-node Kubernetes cluster on Proxmox (kubeadm, not k3s), Jenkins CI/CD platform, dynamic agent execution

Skills Added — Platform Engineering Phase

- Git workflow management — branching, commits, repository lifecycle management
- Repository design and standardization across eleven platform projects
- Technical documentation — READMEs, architecture descriptions, technology stack documentation
- Architecture visualization — platform relationship mapping and diagram creation
- Portfolio standardization — consistent naming conventions, structure, and professional presentation
- Validation reporting — play outcomes, test results, and lessons learned documentation
- Operational runbook creation — recovery procedures formally documented
- Architecture Decision Records (ADRs) — engineering decisions documented and version-controlled
- Platform lifecycle management — version control across active and completed projects

Platform Engineering Portfolio — github.com/luisorlandosolis

Eleven platforms, each documented and version-controlled on GitHub, covering everything from secure access to a live production environment and a Kubernetes-native CI/CD delivery platform.

Dev-Ops-01 Secure Remote Operations Platform ✓ Complete

Stack: Azure · Terraform · WireGuard · Grafana · Prometheus · DDNS · Nginx · Bastion

- Secure hybrid administration through Azure and on-premises environments via encrypted VPN
- WireGuard mesh VPN — encrypted tunnel across all nodes, hardened and locked configuration
- Bastion-based operations management — SSH keyless auth, interactive menu, VPN-gated access
- Terraform-managed Azure infrastructure — on-demand public IP with DDNS integration
- Grafana + Prometheus — NOC dashboard, real-time WireGuard peer monitoring, FinOps views
- FinOps: on-demand public IP model — pay only when needed, zero charges on teardown
- Serial console recovery workflow — device-code login → Terraform apply → SSH in minutes

Dev-Ops-02 Multi-OS Automation Platform ✓ Complete

Stack: Ansible · Fleet OSS · Orbit · osquery · SSH · WinRM · Ansible Vault · Docker · MySQL · Redis · Go

- Centralized Linux, Windows, and macOS lifecycle management — agentless via Ansible
- Fleet OSS MDM — 10 hosts enrolled across macOS, Windows Server, Windows 11, Ubuntu, Rocky Linux, Fedora
- INI-driven inventory — Linux via SSH, Windows via WinRM/NTLM, macOS via SSH — one playbook, all platforms
- Cross-compiled agent binary deployment per platform from a single Linux build host
- Universal Ansible playbooks — single runtime variable targets any inventory group at scale
- Ansible Vault — zero plaintext secrets across all playbooks and inventory
- Idempotent deployment — same playbook handles fresh install, existing system, or broken state
- FinOps: Fleet OSS eliminates per-device MDM licensing costs entirely

Dev-Ops-03 Hybrid Infrastructure Provisioning Platform ✓ Complete

Stack: Terraform · Ansible · Azure · Proxmox · Active Directory · WireGuard · Hyper-V · Python

- Infrastructure as Code — Terraform provisions Azure VMs, VNet, subnets, NICs, and public IPs
- Automated destroy-and-rebuild pipeline — full Azure environment rebuilt from scratch in one command
- Terraform → Ansible auto-handoff — VPN mesh configured immediately post-provision, zero manual steps
- Backup and restore workflow — user data survives a full machine teardown and rebuild cycle
- Active Directory integration — domain enrollment automated through the provisioning workflow
- Zero-touch provisioning — machine joins domain, VPN mesh, and receives full configuration automatically
- Proxmox 3-node HA cluster — sample application with automatic failover across cloud and on-premises
- Domain controller replica — FSMO roles transferred, two Domain Controllers with AD DS and DNS operational

Dev-Ops-04 FinOps & Cost Observability Platform ✓ Complete

Stack: Python · PostgreSQL · Grafana · Azure Cost Management API · SQL · Ansible · Cron · Bash · Docker

- Automated Azure cost ingestion — Python pipeline pulls data from Azure Cost Management API on a schedule
- PostgreSQL storage — structured cost data with SQL-based reporting and historical trend analysis
- Grafana FinOps dashboards — cost governance views, trend analysis, resource consumption visibility

- Ansible + cron-scheduled runs — dashboards stay current automatically without manual intervention
- Cost-driven infrastructure decisions demonstrated across licensing and IP allocation choices
- Azure VM shutdown automation — scheduled lifecycle management reduces idle resource costs

Dev-Ops-05 Certificate Lifecycle & Security Response Platform ✓ Complete

Stack: OpenSSL · Ansible · Ansible Vault · Flask · Docker · Prometheus · Grafana · Nginx · Python · WinRM · SSH

- Internal certificate authority — full control over trust, identity, and certificate issuance across all nodes
- mTLS enforcement — staged rollout: TLS validation → optional client cert → full zero-trust enforcement
- Cross-platform — unified zero-trust model spanning Linux and Windows certificate stores
- Time-boxed certificate lifecycle with automated renewal via scheduled jobs
- Grafana Security Command Center — certificate expiry countdown, health status, one-click breach simulation and remediation
- Flask-based security automation API — webhook receives Grafana alerts and triggers Ansible playbooks
- Full incident response cycle — breach to full remediation across all nodes in under 1 minute
- Security architecture designed following Zero Trust and defense-in-depth principles
- Email notifications at each stage — breach detected, remediation triggered, completion confirmed

Dev-Ops-06 High Availability & Recovery Automation Platform ✓ Complete

Stack: Ansible · Flask · Docker · WireGuard · Fleet OSS · Grafana · Prometheus · Proxmox · Azure · Python · Rsync · Nginx

- Inspired by a real-world power outage — automates recovery steps that were previously performed manually
- 11 recovery plays fully implemented and validated across hybrid on-premises and Azure environment
- Automated service health checks, restart, and notification across the environment
- Tiered critical-service recovery sequence — core connectivity restored before dependent applications
- Backup and cron validation — retention policies, trust remediation, and backup execution confirmed automatically
- Monitoring and warning webhook pipeline — event detected → automated response → notification
- Disaster recovery validation — backup coverage, recovery points, and DR readiness assessed automatically
- Documented, repeatable post-outage recovery sequence
- Azure regional recovery node — cloud DR target deployed and validated
- Structured logging, state tracking, and duplicate alert suppression throughout

Dev-Ops-07 Solis Enterprise Platform (Kraken) ✓ Complete

Stack: Linux Administration · SSH · Configuration Management · Automation Workflows · Service Integration · Infrastructure Documentation

- Built and validated a centralized infrastructure management environment supporting secure remote administration, automation, monitoring, and operational standardization across multiple systems
- Applied enterprise DevOps practices — Linux administration, SSH-based remote management, configuration management, automation workflows, infrastructure documentation, and service integration
- Built with room to grow, without giving up security or repeatability
- Delivered server deployment, service integration, automation development, operational validation, and documentation standardization
- Leans on the same fundamentals as the rest of the portfolio: infrastructure as code, consistency, and centralized administration

Dev-Ops-08 Backup & Recovery Automation Platform ✓ Complete

Stack: Proxmox VM Backup · Replication · Archival Storage · Cloud Sync · Windows & Linux Protection · Retention Policy Automation

- Built a multi-tier backup architecture protecting critical infrastructure, virtual machines, application data, and configuration repositories
- Implemented a layered recovery strategy — primary backup repository, replicated storage, archival storage, and cloud-based offsite protection
- Created and validated automated recovery jobs to move backups across storage tiers while maintaining retention policies and disaster recovery objectives
- Covered Proxmox VM backups, configuration backups, replication workflows, Windows and Linux system protection, archive management, and cloud synchronization
- Tested to confirm it actually holds up, with multiple recovery points spread across separate storage locations

Dev-Ops-09 Security Station Deployment & Recovery Platform ★ Live Production

Stack: Hyper-V · Clonezilla · PowerShell · Ansible · WinRM · Ansible Vault

- Automates Security Station deployment, recovery, and lifecycle management in a live production environment — replacing lengthy manual recovery procedures with a repeatable deployment process
- Validated automated deployment and configuration of Security Station virtual machines using Hyper-V, Clonezilla recovery workflows, PowerShell automation, Ansible configuration management, and WinRM remote administration
- Delivered parent image recovery, child VM deployment, domain integration testing, kiosk validation, barcode scanner support, and label printer compatibility
- Built a technician-focused deployment GUI, plus automated validation and recovery workflows to simplify operational support and maintain consistency across deployments
- Roadmap includes monitoring platform integration, advanced automation frameworks, disaster recovery orchestration, and further infrastructure standardization

Dev-Ops-10 Kubernetes Platform Engineering & Operations Platform ✓ Complete

Stack: Kubernetes · kubeadm · containerd · Flannel · CoreDNS · NGINX Ingress · MetalLB · SMB CSI · Jenkins · Proxmox · Ubuntu 24.04

- 3-node Kubernetes cluster built from scratch using kubeadm on Proxmox VMs — not k3s, not a managed service, full control plane ownership
- Control plane + 2 worker nodes — containerd runtime, Flannel CNI networking, CoreDNS service discovery
- NGINX Ingress Controller, MetalLB load balancing, SMB CSI persistent storage — production-inspired platform architecture
- Jenkins deployed on Kubernetes with SMB-backed persistent storage, HTTPS access via Nginx Ingress, and GitHub integration
- Static Jenkins agent for infrastructure automation (Terraform, Ansible) — SSH-based, validated build execution
- Dynamic Kubernetes agent execution — Jenkins provisions disposable agent pods on demand, automatic cleanup after build
- RBAC + ServiceAccount authentication — cluster-level authorization model validated
- Validated by running the Azure FinOps Cost Observability platform through a dynamically provisioned Kubernetes agent — real workload, not a demo
- Namespace isolation, pod self-healing, replica scaling, persistent volume claims — all validated
- Architecture Decision Records (ADRs) committed to repository — engineering decisions documented and version-controlled
- Harbor Container Registry, GitOps with ArgoCD, advanced observability — planned next phases

Dev-Ops-10.5 CI/CD Platform ✓ Complete

Stack: Jenkins · GitHub · Kubernetes · Pipeline-as-Code · Groovy · Bash · YAML · SMB CSI · NGINX Ingress · HTTPS · SSH · WinRM

- Companion platform to Dev-Ops-10 — dedicated CI/CD environment for software delivery, pipeline automation, and build orchestration
- Jenkins deployed on Kubernetes with persistent storage and HTTPS — platform-grade CI/CD, not a standalone install
- GitHub integration using Fine-Grained Personal Access Tokens — secure credential management via Jenkins credential store
- Pipeline-as-Code — Jenkinsfiles version-controlled in GitHub, pipeline changes are auditable and automatically executed
- Poll SCM automated build triggering — builds fire automatically on repository changes without exposing Jenkins to the public internet
- Static Jenkins agent for infrastructure automation workloads — Terraform, Ansible, Git operations via SSH
- Dynamic Kubernetes agent execution — disposable pod-based build environments provisioned on demand, cleaned up automatically
- Validated against the real Solis weather platform — Weather Node A, Weather Node B, Azure DR node, and load balancer health all confirmed through pipeline execution
- WireGuard connectivity, hybrid infrastructure, and Azure disaster recovery all validated through automated CI/CD pipelines
- Multi-agent architecture — controller execution, static agent execution, and dynamic Kubernetes agent execution all operational
- Architecture Decision Records committed to repository — why Poll SCM over webhooks, why PAT over deploy keys, all documented

Platform Progression

Each platform builds on the last, forming a complete DevSecOps, SRE, and cloud-native engineering capability stack:

Platform	Capability	Builds On
Dev-Ops-01	Secure access — WireGuard mesh, bastion, Azure Terraform, monitoring	Foundation
Dev-Ops-02	Multi-OS automation — Ansible, Fleet MDM, agentless management	Secure access
Dev-Ops-03	Hybrid provisioning — Terraform IaC, zero-touch domain join	Multi-OS automation
Dev-Ops-04	FinOps & cost observability	Provisioning
Dev-Ops-05	Security engineering — PKI, mTLS, automated incident response	Observability
Dev-Ops-06	HA & recovery automation — 11 validated plays, Azure DR	Security engineering
Dev-Ops-07	Centralized enterprise administration	Automation & observability
Dev-Ops-08	Backup architecture & disaster recovery	HA & recovery automation
Dev-Ops-09	Live production deployment & recovery automation	Multi-OS + backup/recovery
Dev-Ops-10	Kubernetes platform engineering — kubeadm, Jenkins, dynamic agents	Full infrastructure stack

Dev-Ops-10.5	CI/CD delivery platform — Pipeline-as-Code, static + dynamic agents	Kubernetes platform
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Real-World Incident Response — June 2026

3-day power outage · Full recovery in 2.5 hours · Unassisted · No runbook

What Failed Across 10+ Nodes

- Filesystem corruption on a Linux node — unclean shutdown corrupted the journal
- Invalid fstab UUID entries — system dropping into emergency mode on every boot attempt
- Root filesystem mounting read-only — all write operations blocked across the OS
- Firewall masquerade disabled — WireGuard inter-peer routing broken across the entire mesh
- Fleet Orbit agent failing — missing systemd service flags across multiple nodes
- FinOps Azure cost ingestion pipeline broken — missing environment variable sourcing
- SSH agent state lost across the bastion and all client machines

Recovery Sequence Executed

- Live GRUB edit to bypass systemd and gain write access to the root volume
- Filesystem repair against the corrupted, read-only root volume
- Corrected invalid fstab UUID entries for boot partitions
- Restored firewall masquerade and trusted-zone configuration for the VPN interface
- Corrected systemd unit files and certificate path mismatches for the endpoint agent
- Repaired the FinOps ingest script's environment variable sourcing, resolving a database auth failure
- Restarted WireGuard on the domain controller after confirming it failed to auto-recover
- Reloaded the SSH agent across the bastion and all affected client machines

Post-Incident Hardening Applied

- Updated container restart policies to auto-recover after host reboot
- Hardened bastion shell profile for persistent SSH agent and key auto-load on login
- Captured a verified-good snapshot before the final reboot
- Corrected and documented systemd unit files across all Linux nodes
- Updated the FinOps ingest script to always source required environment variables

What This Became

Every manual recovery step from that outage later became an automated play in Dev-Ops-06, so the same recovery that took 2.5 hours by hand now runs automatically — detected, remediated, validated, and reported without anyone needing to step in.

On-Demand Azure Public IP — FinOps Terraform Project

A Terraform project that creates and destroys an Azure public IP on demand — eliminating the cost of a static IP when not needed. Infrastructure cost governance applied at the resource level.

- Running terraform apply creates the public IP, attaches it to the NIC, and DDNS auto-updates within minutes
- Running terraform destroy detaches and deletes the IP, with zero ongoing charges
- Full lifecycle managed through dedicated Terraform resources, including automatic cleanup of stale IPs
- A scheduled job on the Azure VM keeps the public DNS record current automatically
- Serial console recovery path — Azure portal → Serial Console → login → terraform apply → SSH access restored in minutes

FinOps impact: Standard static public IPs in Azure accrue charges continuously once allocated. On-demand provisioning means the IP exists only when needed — cost drops to zero between sessions.

Technical Skills

Category	Skills
Platform Engineering	Git workflow, repository design, README documentation, architecture visualization, ADRs, portfolio standardization, validation reporting, runbook creation, platform lifecycle management
Networking	WireGuard VPN mesh, subnetting, NAT, firewall management, DNS, load balancing, Nginx reverse proxy, DDNS, virtual switch management
Identity & Security	Active Directory, FSMO roles, DC replication, Kerberos authentication, disk encryption, SSH keyless auth, bastion architecture, mTLS enforcement, Zero Trust
Systems Admin	Windows Server, Fedora, Ubuntu, Rocky Linux, Proxmox cluster management, Samba, cross-platform integration, filesystem repair, GRUB recovery, systemd
Cloud Engineering	Azure VNet, VM lifecycle, Terraform IaC, on-demand resource provisioning, Azure Cost Management API, regional disaster recovery, hybrid cloud architecture
Kubernetes & Container Orchestration	kubeadm cluster deployment (not k3s), containerd runtime, Flannel CNI, CoreDNS, NGINX Ingress, MetalLB, SMB CSI storage, namespace isolation, RBAC, pod self-healing, replica scaling, persistent volumes
CI/CD & Software Delivery	Jenkins on Kubernetes, Pipeline-as-Code (Jenkinsfile), GitHub integration, Poll SCM automation, static agent execution, dynamic Kubernetes agent execution, multi-agent build architecture, credential management
DevOps & Automation	Terraform (destroy/rebuild pipelines), Ansible (agentless, universal playbooks), Docker, cron, auto-provisioning, webhook automation
Security Engineering	Internal PKI, staged mTLS rollout, Zero Trust, certificate lifecycle management, Ansible Vault, breach simulation, closed-loop remediation, Flask security APIs
Monitoring & FinOps	Prometheus, Grafana NOC + Security Command Center, certificate expiry alerting, Azure cost ingestion, PostgreSQL, SQL reporting, event-driven remediation, FinOps governance
Endpoint Management	Fleet OSS MDM, osquery telemetry, multi-OS enrollment, agent deployment, Windows service lifecycle, macOS agent management
SRE & Resilience	Incident response, filesystem recovery, backup auditing, DR validation, tiered service recovery, recovery automation, Azure regional DR, structured logging
Enterprise & Production Ops	Centralized administration platforms, live production deployment automation, multi-tier backup architecture, Clonezilla imaging, kiosk

	deployment automation, technician-facing tooling
Development	Python (cost ingestion, applications, Flask APIs), PowerShell, Groovy (Jenkinsfiles), Bash, SQL, application caching, systemd service management

What's Done, What's Next

Platform	Status
Dev-Ops-01 Secure Remote Operations	Done
Dev-Ops-02 Multi-OS Automation	Done
Dev-Ops-03 Hybrid Provisioning	Done
Dev-Ops-04 FinOps & Observability	Done
Dev-Ops-05 Certificate & Security Response	Done
Dev-Ops-06 HA & Recovery Automation	Done
Dev-Ops-07 Solis Enterprise Platform (Kraken)	Done
Dev-Ops-08 Backup & Recovery Automation	Done
Dev-Ops-09 Security Station Deployment & Recovery	Done — Live Production
Dev-Ops-10 Kubernetes Platform Engineering	Done
Dev-Ops-10.5 CI/CD Platform	Done
Azure Policies / IaaS-SaaS Governance	In Progress
MFA + Microsoft 365 Deep Integration	In Progress
Harbor Container Registry + GitOps (ArgoCD)	In Progress
SIEM Integration	In Progress
UPS + Power Management Integration	In Progress
FinOps Capstone Platform	Planned

Key Problems Solved

- WireGuard peer loss — persistent-config setting silently overwriting live configuration on restart; corrected and locked down
- Linux domain join failure — incorrect hostname resolution; corrected, Kerberos authentication completed successfully
- Hyper-V network hijacking — virtual switch competing for the default route; isolated network interfaces, resolved permanently
- VPN routing failures — overly broad allowed-IP ranges; corrected all peers to strict point-to-point routing
- Firewall zone conflicts — VPN interface assigned to multiple conflicting zones; rebuilt cleanly with single-zone architecture
- Certificate breach simulation — full automated incident response: key destruction, revocation, reissuance, and redeployment via Ansible
- Endpoint agent enrollment failure — service lifecycle race condition; full reset workflow engineered, permissions issue solved

- Endpoint agent TLS chain failure — multiple certificate and permission mismatches across the trust chain, all identified and resolved
- Filesystem corruption recovery — live bootloader edit, filesystem repair, and configuration correction completed unassisted
- FinOps pipeline failure — missing environment variable sourcing in the ingest script; corrected, database authentication restored
- SSH trust mismatch — host key conflict breaking backup workflows; remediated, backup execution restored
- SMB CSI + Jenkins SSH key permissions — CIFS-backed storage introduced OpenSSH temporary key file permission challenges; solved by switching to GitHub Fine-Grained PATs
- Kubernetes SCM webhook limitation — private infrastructure limits GitHub-hosted webhooks; Poll SCM implemented as reliable automated trigger alternative

Portfolio Summary

This portfolio reflects hands-on work across platform engineering, DevOps and DevSecOps practices, cloud infrastructure, site reliability, systems administration, Kubernetes platform engineering, CI/CD delivery, and hybrid cloud architecture — including one platform running in a live production environment and one Kubernetes platform validated by executing real operational workloads through dynamically provisioned agent pods.

There are eleven platforms here, each with its own GitHub repository. Dev-Ops-06 grew directly out of a real outage recovered manually and later automated into 11 recovery plays. Dev-Ops-10 was built using kubeadm on bare Proxmox VMs — not k3s, not a managed service — because understanding what Kubernetes actually does under the hood matters. Dev-Ops-10.5 closed the software delivery loop by adding a full CI/CD platform on top of the Kubernetes infrastructure. Each platform is documented and version-controlled, so the code and write-ups are there to review directly.

Currently studying for AZ-104.

GitHub: github.com/luisorlandosolis — eleven platform repositories with documentation, stack descriptions, validation notes, architecture diagrams, and Architecture Decision Records.